

Supplemental Information – – Extended Methods

Computational strategy to build PhosphoAtlas database

We developed a computational strategy for building a centralized and harmonized relational database where human protein kinases are associated with their phospho-target substrates and validated residue target sites and heptameric (phospho-)peptide sequences (HPS). A series of methods and systems allows to logically build such database from both kinase-centric and substrate-centric perspectives, eventually converging toward the identification of all existing, verified, functional relationship between a kinase and a substrate and/or a phosphorylatable catalytic site. **Figures 1** and **S1** depict the core logic of the systematic processing loops of computational steps and regression-like curation systems. Strategic steps to extract, integrate, unify, and harmonize data from public databases (see Phase 1), and to build a relational database including methods to identify, curate and centralize verified, validated, human protein phosphorylation events (see Phase 2) under a unique database are depicted below.

Public databases and related data/publications utilized to build PhosphoAtlas and analyzed employing computational curation methods analogous to and modified from ones described by (1-6), include: APID (7), Bindingdb (8), BioGRID (9), CCDS (10), CleanEx (11), CLIP (12), COSMIC (13), DIP (14), EMBL (15), Ensembl (16), EPD (17), Gencode (18), GO (19), HAVANA (16), HGNC (20), HPRD (21), H-InvDB (22), IPI, iPfam (23), KEGG (24), KinG (25), MEROPS (26), MiMi (27) and UniHI (28), MINT & IntAct & MIntAct (29), MPact, NCBI (30), OMIM (31), PANTHER (32), PDB (33), PhosphoSitePlus (6), PhosphoPOINT (34), Phospho.ELM (35), Pfam (36), PINA (37), PubMed, RCSB (38), SNP (39), STRING (40), Transfac (41), Treefam (42), Trome (43), UniGene, UniProtKB/Swiss-Prot (44), UniProtKB/TrEMBL (44), UCSC (45), UTRdb (46), Vega (47).

In summary, the first phase of the computation supports the extraction and harmonization of public human protein records, mainly to prevent denomination inaccuracies, ambiguities or redundancies, and to enable structuring large amounts of information in a searchable repository. The second phase of the computation enables identification of active components of phosphorylation networks, with a specific set of methods designed to identify kinase and substrate proteins/genes, identify accurate, verified phosphorylatable residues, and finally extract the related biological HPS from protein substrates.

Phase 1 – Protein index curation: Methods to integrate multiple data sources into one harmonized protein index system (related to Figure 1 and S1)

The method starts by creating a reference index. We generated a 'Protein Reference Index' from HGNC and NCBI/Entrez to create a reliable, non-redundant inventory where each human protein was

defined by its standardized name, symbol, and RefSeq-numbers, which together established a 'Primary Identifier' (**Figure S1**, step 1).

Because the Protein Reference Index is created by cross-referencing the protein sequence records from the Entrez database maintained by the National Center for Biotechnology Information (30), with the standardized gene nomenclature database from the HGNC database maintained by the HUGO Gene Nomenclature Committee, this step ensured proper identification/nomenclature for all later steps of the computational process. The HGNC database contains reliable and standardized gene names along with standardized gene symbol for each known human gene, while each NCBI/Entrez Gene ID in the Entrez database is associated with a Reference Sequence (RefSeq) nucleotide and protein record. The main reliable feature of the NCBI/RefSeq collection includes non-redundancy, explicitly linked nucleotide and protein sequences, updates to reflect current knowledge of sequence data and biology, data validation and consistency, a distinct identifier, and the benefit of an ongoing curation by NCBI staff and collaborators, with reviewed records indicated.

For internal curation-and-tracking purposes, a unique, unambiguous, internal database identifier was additionally created for each HGNC-NCBI matched protein record. This was included "within" the 'Primary Identifier' record, along with the minimal set of protein descriptors (i.e. RefSeq protein and nucleotide record and ID, Entrez Gene ID, standardized name/symbols).

Available additional functional data and non-official aliases were also compiled as complementary content, including –but not limited to– information such as known aliases, chromosome location information, CCDS (Consensus CDS Protein Set) ID, Pubmed ID(s), Ensembl ID, OMIM (Online Medelian Inheritance in Man) ID, UniProt ID, GeneRIFS (list of known functions), list of protein-protein interactions (i.e. potential enzyme-substrate relationships), Gene Ontology information (i.e. functions, processes, cellular localization), and so forth. The additional information associated with each record could be maintained directly with the record in the Protein Reference Index (or stored in separate tables or indices based on data type and associated with the appropriate record), via their unique database identifier.

Next, the Protein Reference Index served as a blueprint to systematically integrate, identify and verify remaining or additional protein records extracted from other available bio-medical data sources, which we referred to as 'external' resources (**Figure S1**, step 2; cross-referencing method analogous to (1)). Using regular expression syntax in Perl, the directly matching external records were imported and structured by their common, unique Primary Identifier, and all associated complementary data were extracted. Additional functionality data not previously associated with the record were added to appropriate table(s) or index(es), to be eventually merged and harmonized in later step 4; as described below. If knowledge did not match any Primary Identifier, the external identifiers from the external records were cross-referenced with the UniProt database in Curation 1 (**Figure S1**, step 3). In order to analyze unmatched records, we reformatted the information from UniProtKB database and

structured it by Primary Identifiers. Once an external identifier was successfully cross-referenced and matched with a record from the UniProtKB index, the related external protein record and associated information were updated under the appropriate Primary Identifier. If a primary identifier could not be determined by cross-referencing the UniProt database, the record was flagged and unmatched data were stored in a Protein Watch Index or excluded.

The Protein Watch Index can be used as an 'external database' while concentric repetitive curation steps are taking place either at this stage or later during the curation process, which enables further accumulation of protein annotations over the course of the entire iteration. Also, acquisition, extraction and filtration of multiple data sources can be run in parallel (i.e. at the same time) once resources are formatted to allow curation. In addition to 35 main, external databases used in the previously described curation steps, information mined from additional literature sources such as handbooks, textbooks and web-based resources/websites, can be added to or associated with records in the Reference Index via an identifier such as a PubMed Id or ISBN number. Similar curation systems, annotation process and verification steps mentioned above applied.

To harmonize the content of this initial protein resource, collected records were curated (**Figure S1**, step 4, curation 2). Curation 2 ensures record quality by complementing and modifying information associated with each extracted record, to offer consistent, non-redundant, unambiguous data sets. This includes checking record label annotation, adjusting record taxonomy, and removing or merging data associated with redundant sequences, organizing information in properly merged/organized indexation pools, and configuring data in searchable format. It also includes both computational and manual-intensive processing. In particular, curation 2 was programmed to execute steps comprising the use of a sequence alignment algorithms such as, but not limited to, BLAST (Basic Local Alignment Search Tool) the Smith-Waterman algorithm or pattern matching computer implemented methods (regular expression syntax in Perl) depending on record format or content, to identify incorrect or non-matching sequences or records. For example, all sequence data were compared with the sequence associated with the primary identifier in the reference index (i.e the sequence contained in the Entrez database for a given RefSeq). When a redundant sequence was found, a cross-reference was made with Entrez to verify that the latest sequence was present and then updated if needed. In addition, any non-redundant data between the records was merged into a single record. If the redundant sequences were associated with two separate records, the data associated with both records was merged into a single record and one unique internal database identifier was discarded. If redundant sequences were associated with a single record, the source information for each redundant sequence was retained and extraneous copies of the sequence discarded. In all cases the retained sequence is the latest sequence in Entrez. Similar strategies were devised and used to check, remove, or merge any redundant records across the different annotations

collected from database/resources. These data were accurately consolidated, and used in later data mining and curation steps (see Phase 2 below).

The computational methods from Step 1 to 4 and Curation 1-2 generated a coherent, centralized, human protein database we named 'Harmonized Proteome Index'. The resulting dataset compiled only non-overlapping, non-redundant, current, comprehensive information validated from cross-referencing and cross-verification of known human proteins.

Phase 2 – Phospho-reactome curation: Computational methods to identify kinase – substrate phosphorylation circuits, and to extract phosphorylatable residues and biological HPS (related to Figure 1 and S1)

The following computational and manual curation steps depict how the relational phospho-reactome database was constructed. Data-mining methods were developed to specifically extract and functionally connect together kinase enzymes, substrate proteins, and their phosphorylation residues/sites. To qualify the abundance of data, the review of available knowledge included a team of curators who systematically mined through information at specific, manually intensive curation steps. Beyond being an *in silico* protein-protein interaction mapping tool made to identify phosphorylation circuits, these methods and resources describe means to identify the critical components of enzymatic events –that is, identifying the exact target residues from substrates that kinases phosphorylate. These methods also depict how to validate and extract phosphorylatable amino acid sequences from substrate proteins and their phospho-residue sites. (Note that such methods would support the systematic analysis of other enzymatically catalyzed, post-translational protein modification reactions (e.g. acetylation, glycosylation, methylation, ubiquitination etc) and map the 'catalytic proteome' in general.)

To establish the map of phosphorylation circuits of human cells, records from the Harmonized Protein Index were first searched using regular expression syntax in Perl and natural language processing (NLP) algorithms (Mergesort and quicksort in Perl, Arrays.sort in Java, timsort in Python) to identify references containing protein kinase enzymes or protein kinase substrates (**Figure S1**, step 5; information garnered from all 38 database resource used in Phase 1 of the method; the scientific literature mining method is analogous to the NLP deployed by other studies in the PPI field, including (6)). At this step, data annotation describing the phosphorylation of amino acids was used as a means to identify candidate proteins that could be considered as kinases or substrates. At this step and the following ones, knowledge of protein modification that logically connects a kinase to a substrate was collected in order to progressively build a curated network of specific kinase–substrate catalytic interactions, but also accumulate evidences supporting the identification of which target sites are phosphorylated by which kinases. The selected records were used to generate an initial kinase and substrate dataset that was next examined via a curation sequence (**Figure 1**, Curation 3, Step 6).

We next developed a target validation procedure (Curation 3, Step 6) based on concentric computational steps that identified exact phosphorylated residues/sequences, and distinguished validated phospho-target candidates, to ultimately create a Phospho-Residue Site Index and a biological Heptameric Peptide Sequence Index logically interconnected to their Substrate Protein Index and Kinase Protein Index (**Figure S1**). Briefly, protein records were systematically checked for phosphorylation modification information. If phosphorylation position information was not available in the literature, the record was added to the substrate index. When a phospho -residue or -site or -peptide sequence was available from functional annotations in the Harmonized Proteome Index and literature/PubMed verification, it was compared to the latest corresponding curated protein sequence (protein RefSeq / NCBI / UniProtKB) using string-mining principles such as Smith-Waterman-derived algorithms (e.g. Basic Local Alignment Search Tool (BLAST)), or pattern matching computer-implemented methods (e.g. regular expression syntax in Perl) (**Figure S1**, curation 3). For example, if a target record listed the site of phosphorylation at the serine found at position 144, the method verified that a serine existed at site 144 in the curated sequence. If the alignment was conclusive, the amino acid sequence surrounding the validated phospho-site was extracted from the protein RefSeq, and the heptameric peptide was indexed, annotated and functionally assigned to its substrate and its kinase(s) (when/if known and verifiable).

However, if the described phosphorylation position, residue or peptide composition could not be confirmed, a warning notification was issued and an optimal alignment against the official RefSeq sequence was generated to identify possible misalignments or misspellings, and determine whether a reasonable adjustment or concordance could be achieved. For example, in cases where records provide peptide information, that is information regarding the composition of amino acids flanking the phosphorylation site, the peptide was aligned with the latest available curated sequence to determine if both the amino acid composition and site of modification could match the curated sequence. Importantly, this step relied on dynamic sequence scanning and alignment algorithms combined with manually intensive verification, but deliberately did not rely on confidence-based tools (commonly used in highly automated computation or high throughput proteomic studies, such as confidence-based scoring approaches, predictive algorithms, conditional selection, correlative methods, conservation analysis or inferred relationships). Such approach would pick many additional but likely partly erroneous peptide sequences/target sites that would then inevitably undermine the accuracy of the entire database. This rigorous filtering process lead to a restricted number of kinase–substrate catalytic interactions available in PhosphoAtlas dataset due to the exclusion of PPI's that could not be cross-validated. When a sequence alignment was judged acceptable, phospho-residue sites and HPS's were extracted from the RefSeq and indexed, whereas when inconclusive, phospho-target site details were eliminated, even though only the original, curated kinase–substrate protein information and functional connectivity remained available.

To further prevent inclusion of unconfirmed data and thus ensure the collection of only validated information, primary records providing candidate targets that could not be cross-referenced via multiple publications/database references using complementary techniques/approaches were excluded. Moreover, we computationally distinguished validated phospho-targets with or without identified upstream kinases, and sub-classified separately ‘orphan’ substrates and (phospho-) peptides. This process resulted in exclusion of a large number of residue target sites and amino acid sequences from the resulting dataset itself; only phosphorylatable substrate proteins with identified upstream kinases are included.

Lastly, all knowledge derived from Curation 3 was logically assigned to four interrelated indexes named Kinase Protein, Substrate Protein, Phospho-Residue Site, and Heptameric Peptide Sequence (HPS), using logical principles similar to ones earlier described in Curation 2 (**Figure S1**, Step 7). Data was stored under one database named ‘PhosphoAtlas’, and this ended the data-mining process. Ultimately, this systematic, manually intensive approach generated a coherent map of interconnected proteins established from concerted sources that compose the verifiable human phospho-reactome.

PhosphoAtlas database and utility

Users may access and explore PhosphoAtlas dataset upon request. Some tabulations in PhosphoAtlas may enable users to programmatically integrate their experimental data onto the pre-build network of kinases, substrates and target sites. As discussed in this study, examples of such applications include: (1) Tissue-, or disease-, sub-cellular compartment- specific studies, since partitioning kinase-substrate pathways depending on protein tissue distribution, subcellular localization, and biological processes as collected in GO, would generate maps that are representative of biological context; (2) Analysis of phospho-proteomic data generated from high throughput proteomic approaches such as mass spectrometry (48-61), or other kinome exploratory means (62-67); (3) Analysis of information about catalytic PPIs using specific computational tools (e.g. SIFT, Provean or SIB/ExPASy/PROSITE (68-70) or other interfaces or conformational analysis tools (71-73)) in order to compare wild-type target regions and their variant counterparts in structure-guided studies of catalytic pockets and accessory or regulatory domains, which could help rational drug design to tailor compounds specific to the mutation profile of diseases.

Supplemental Information – Supplementary Figure Legends

Supplement Figure S1. Phospho-reactome curation workflow.

Resources, computational processes, and protein/peptide repository are presented (see Extended Methods for detailed explanations). The method relies on a series of concentric curation systems that correct and complement records, as data is being extracted and brought in under consistently annotated indexes. This routine progresses through pre-established building blocks and selection steps, using analogous computational and manually intensive curation tools to filter and merge data. Further analysis and unification systems enable identification of latest available, validated, exact phospho-residue target sites and amino acid sequences targeted by kinase enzymes, and exclude ambiguous records or probability-based-only records. PhosphoAtlas contains only one, non-redundant, correct set of information for each molecule across multiple protein classes, while functionally connecting proteins based on their biochemical catalytic interaction(s). Step 1 to 4 and Curation 1 and 2 support the extraction, integration and harmonization of public protein records, mainly to prevent denomination inaccuracies or redundancies, and to enable structuring large amounts of information in a searchable repository (named 'Harmonized Proteome Index'). Step 5 to 7 and Curation 3, allow identification of active components of phosphorylation networks, with a specific set of methods designed to identify exact, verified phosphorylatable residues, and extract the related biological peptide target sequences from their cognate protein substrates.

Supplement Figure S2. Proteins catalogued in PhosphoAtlas correspond to a unique subset of Protein-Protein Interactions (PPI), defined by their active, proven function in phospho-catalytic events.

The hairball plot represents the total number (17,109) of human Protein-Protein Interactions (74) established by the PINA2 data resource (all dots) (1). Proteins identified in PhosphoAtlas as verified effectors of phospho-circuits are highlighted in color: kinases are in blue; substrates are in red; and proteins that are both a kinase and a substrate are shown in purple. PhosphoAtlas contains 11.25% of all PPI nodes defined in the PINA2 database that are verifiable and curated using PhosphoAtlas computational pipeline. Representation does not include the detail of phospho-catalytic edges related to phospho-residue sites and HPS's. This representation also does not include substrates that are not yet conclusively mapped by PhosphoAtlas to a kinase.

Supplement Figure S3. Comparative connectivity between kinases, substrates and HPS's.

a-c. The network of phosphorylation connections for **(a)** kinases and **(b)** substrates is displayed based on the number of connections between unique peptide targets either found in **(a)** their downstream substrates or **(b)** phosphorylated by their upstream kinases. The x-axis represents **(a)** the number of

substrates per kinase, or **(b)** the number of kinases per substrate. The y-axis represents the number of distinct peptide sequence **(a)** per kinase or **(b)** per substrate. Graph **(c)** presents the connectivity of single HPS's interconnected with kinase(s) (x-axis) and substrate(s) (y-axis). For representation purposes, overlapping individual discs with identical (x;y) values were randomly, incrementally offset in **b,c** graphics.

d. The total number of distinct peptide sequences in the database was counted depending on the pre-defined chosen length of peptides extracted from their protein substrate of origin (either 7 or 9 or 11 or 13 amino acids in length, and centered on the phosphorylatable residue). The graph shows that there is only marginal gain in 'specificity' from using peptide sequences longer than 7 amino acids, as observed in various biochemical and biological contexts (75, 76).

Supplement Figure S4. Most highly connected kinases and substrates.

a-b. Top 100, most connected kinases and substrates. Waterfall plot connecting substrates **(a)** or kinases **(b)** to their enzymatic partner (blue bars). The number of known HPS's per kinase or per substrate is shown as red bars on top of each blue bar.

c. Top 50 kinases for which the largest difference between the number of substrate (red bars) and HPS's (blue bars) is shown. Kinases are organized as a waterfall plot of highest to lowest number of HPS's per kinase.

Supplement Figure S5. Comparison between the number of enzyme – substrate pairs and the number of kinase – HPS – substrate connections across kinase subfamilies.

The graph in **(a)** shows the number of kinase – substrate pairs per sub-family of kinase. The top 9 subfamilies with ≥ 100 kinase-substrate pairs are highlighted in color. The graph in **(b)** shows the number of kinase – HPS – substrate triples (i.e. catalytic edges) per subfamily of kinase enzyme. Circos plots of the top subfamilies shown in **(a)** and **(b)** serve as visual representation of the complexity of the phospho-catalytic interactivity. In each circos plot, proteins and peptide sequences are sorted by their alphabetical name within their respective group. Comparing the level of connectivity from these two graphs reveals how adding HPRs/HPS's information unveils the otherwise hidden complexity of phosphorylation networks. For these graphs, only kinase subfamilies including kinases with identified substrate targets are listed. We classified all human kinase genes identified from our curation process into 128 kinase sub-families using classification system defined by Manning et al, Science 2002 (77) alternatively, kinase genes could be catalogued into 373 identifiable family subclasses using Pfam modalities (36), but this was not used here for clarity purposes). Kinase family names are based on publication by Manning et al. (77) and correspond to: AGC: contains PKA, PKG, PKC families; TK: Tyrosine Kinase families; CMGC: contains CDK, MAPK, GSK3, CLK families;

CAMK: calcium/cadmodulin-dependent protein kinase; STE: homologs of yeast sterile 7/11/20 kinases; CK1: casein kinase 1; TKL: tyrosine kinase-like; RGC: receptor guanylate cyclases.

Supplement Figure S6. Disease ontology profile of AKT phospho-circuits.

- a. Disease Ontology (DO) terms were classified in 13 distinctive categories (column 2). The number of unique proteins from the AKT network per each category (column 3) is subsequently shown.
- b. Per AKT network protein, the number of DO terms per disease category is given.

Supplement Figure S7. Connectivity of AKT proteins to their upstream and downstream phospho-protein circuits.

Hive plot (78, 79) network representation of known interactions of AKT1, AKT2 and AKT3 proteins (top to bottom respectively). Hive plot visualizations of AKT family proteins, which can act as either kinases or are the phosphorylated substrates, show the number of substrates and kinase enzymes interacting with each protein, along with the proportion of known peptide targets. Every axis contains all alphabetically ordered kinases (red), substrates (blue) or peptides (green), as stored in PhosphoAtlas (i.e. every unique node of two interconnected entries are mapped to and positioned on radially distributed linear axes). The grey connecting curved links (or edges) represent known phosphorylation events when a residue target site and related peptide sequence from a given substrate is known, and thus any grey line goes through the peptide axis before reaching its cognate substrate. The gold edges depict phosphorylation events when there is no known or verifiable target site reported in the literature, thus connecting the kinase directly to the substrate. Each of the nine panels is annotated with the number of unique kinases, substrates, or peptide sequences represented in the network as well as the total number of edges, which is the total number of known connections between kinases, substrates and the target peptides. Hive plots on the far left show how AKTs connect with their upstream kinases. Hive plots in the middle part show how AKTs connect with their downstream substrates (and peptide target sequences). Hive plots on the right display the merged upstream and downstream connectivity of each AKT (i.e. AKT-centric phospho-network connectivity).

Supplement Figure S8. Catalytic circuitry of AKT1 kinase: an expandable network of enzymatic interaction derived from mutation-prone phospho-target sites.

- a. Number of peptide target sites per substrate protein and phosphorylated by AKT1 kinase. About one third of protein substrates have yet to be associated with a definitive residue target site. For about two third of AKT1 substrates, residue target sites and HPS targets have been identified (although more target sites may be found or verified for some of these proteins during future iterations of the computational process). Externally collected genomic mutations were queried to identify whether they encoded a non-synonymous alteration in the amino acid sequence of peptide target sites (named:

non-synonymous Heptameric Peptide Sequence variant, or 'nsHPRv'). The number of additional (non-synonymous variant) peptide sequences is plotted (tan-colored bar) on top of the original number of individual peptide target sequences (grey bar).

b. Overview of all possible genetic variants affecting the AKT1 circuit. For example, the first top left triangle shows that one peptide target site of AKT1 can be altered in 9 different ways (i.e. y-axis value = 9), and these variants are not single nucleotide polymorphisms (SNPs) since the intense blue color of the triangle corresponds to a value of 9 on the non-SNP versus SNP ratio color scale. Two other target sites within the AKT1 protein are affected by variants (2 deeper-blue triangles below); and two more peptide target sites of the AKT1 protein are not affected by any known genomic variations (visible as 2 overlapping grey triangles at level '0' on the y-axis; the visible triangle is darker grey due to the overlap of the two grey triangles, which also correspond to a '0' value of the non-SNP/SNP ratio score from the color scale). Many of the proteins from the AKT1 network (but not necessarily targeted at all their phospho-residue sites by AKT1) may be affected by alterations within the 3-amino acid-vicinity of phospho-catalytic sites. For some of these proteins, all of their residue target sites are potentially subjected to altering mutations (e.g. BRAF, PTEN). Most of these alterations are not driven by SNPs (i.e. germline mutations affecting >1% of the human population), but found at lower frequency in diseased tissues.

c. 52 out of the 85 peptide target sites found across the 56 substrates phosphorylated by AKT1, are affected by variants that, in total, correspond to 91 additional, altered target sequences. Out of the 91 variants, 85 are single amino acid substitutions.

d. Biological peptide target alignment using all 85 unique 7-mer sequences phosphorylated by AKT1 enabled comparing preferred amino acid sequences, and to establish which residue are most often conserved within the vicinity of the catalytic sites recognized and phosphorylated by AKT1. Consensus sequences were made using: <http://weblogo.berkeley.edu/logo.cgi>. More globally, it is important to notice that there is no evidence for further conservation within the target site region when one takes into account that no HPS's are shared between substrates targeted by the same kinase.

e. Absolute frequency of mutations found across peptide variants are shown. Out of the 7 amino acids, the Arginine in position (-3) was the most disproportionally affected by non-synonymous variations. (i) The frequency of loss of Ser(0) or Thr(0) is slightly lower than the 'predictable' mutation rate of residues within a HPS, assuming that each residue site and each amino acid has the same chance to be altered (i.e. 1 out of 7 $\approx$ 14.3%, in comparison to 10 peptides with a loss of Ser(0) or Thr(0) out of 85 variant peptides $\approx$ 11.8%). (ii) Loss of Arg(-3) has $\sim$ 2.5 times higher chance than 'normal' to be altered (i.e. expected $(1/7) \times (44 \text{ peptides with an Arg(-3) out of } 52) \approx 12.1\%$, in comparison to (26 peptides with a loss of Arg(-3) out of 85 variant peptides) $\approx 30.6\%$). (iii) Loss of Pro(+2) has a $\sim$ 3 times higher chance than 'normal' to be altered (i.e. expected $(1/7) \times (10 \text{ peptides$

with a Pro(+2) out of 52) $\approx 2.7\%$, in comparison to (7 peptides with a loss of Pro(+2) out of 85 variant peptides) $\approx 8.2\%$).

Supplement Figure S9. Identification of substrate proteins that are most affected by cancer-mutated peptides, and substrates that are not subject to alterations of their phospho-target sites.

a-c. Extent of non-synonymous variants across nodes (proteins), edges (kinase-substrate) and HPS's (defining the edges). Panel (c) serves as an overview of the impact of genomic mutations on phosphorylatable heptameric regions (HPR). This further indicates that high mutational susceptibility a particular HPR around a target site (**Figure 5e**) is not random and relates to its selective significance for tumorigenesis, hence making it a potential hallmark of cancer-related molecular expression.

d-e. Overview of degree of non-synonymous cancer mutations per protein. In (d), each substrate protein collected in PhosphoAtlas was analyzed for its number of known peptides (x-axis) versus its mean number of total variants per peptide (y-axis) (see **Figure 5g**). The distribution is categorized in boxes (A, B, C, D) for which numbers are detailed in tables in (e). Each substrate is color-coded by the mean percentage of non-synonymous variants per total number of variants in a given substrate protein. This spectrum also identifies a number of substrates where phosphorylation target sites have had no variants reported (box D). In (e), tables with examples of how substrate target sites can be affected by variants caused by genetic mutations found in tumor tissues. Considering every peptide-altering variant (resulting from distinct substitution, deletion or insertion) created one new peptide, and that every peptide-altering variant contained only one amino acid change per HPS, this generated a new group of 2,896 mutated peptides. Out of the original 2,617 HPR's, 1,031 are not affected by any known mutation, 217 contain only silent variants (synonymous) with no effect on amino acid composition, and 1,369 contain at least one non-synonymous variant, out of which 377 include both synonymous and non-synonymous variants, and 992 contain only amino acid-altering variants.

Supplement Figure S10. Distribution and classification of SNP-altering target sites.

a. Distribution of variants of heptameric peptide target sites reported as somatic mutations or single nucleotide polymorphisms (SNPs).

b. Spectrum of substrate proteins affected by SNP variants that alter their heptameric peptide target sites. All substrates from PhosphoAtlas are distributed by the mean number of total variants per HPR versus the number of known peptides per given substrate. Each substrate is color-coded by the mean percentage of SNP variants per total number of variants in a given substrate. The complementary panel underneath the main graph contains only substrates for which no variants in phospho-target peptide sites have ever been reported.

c. Substrate proteins with peptide targets that have the highest proportion of reported SNPs. The top 50 peptides with the highest proportion of SNP variants to total number of variants per peptide were first identified, then the corresponding substrate proteins were extracted, and finally all peptide target sites from these substrates were included on the graphic (including peptide target sites that are not affected by any SNP-related variants, such as the CDC25C's HPR at phospho-residue site S216, which corresponds to the RSPSMPE sequence). The 42 identified substrate proteins are sorted alphabetically on the x-axis. Each peptide is represented as a triangle on the scatterplot, and color-coded by the mean percentage of SNP variants out of total reported variants in a given peptide.

d. The ratio of non-SNP to SNP-associated variants is shown for substrates with the most mutable peptide sites across all human cancers (see **Figure 5h**).

Supplement Figure S11. Heptameric target variants affecting phosphorylation sites mapped to COSMIC-curated tissue pathologies.

Specific cancer pathology information provided by the COSMIC database associated with uniquely mutated peptides derived from PhosphoAtlas peptides indicating which cancer and tissue types are most prone to include mutations that directly affect phosphorylation circuits.

a. A cross section of primary pathological processes (COSMIC-defined) associated with the identified variants reveals that 764 (44.3%) of the alterations are connected to carcinomas, followed by a number of other malignant transformations, such as hematopoietic neoplasms, malignant melanomas, or gliomas.

b-c. Secondary pathology study (COSMIC-defined) further narrows down tissues (b) and malignancy subtypes (c) that PhosphoAtlas peptide variants relate to.

Supplement Figure S12. Identification of kinases impacted by cancer-mutations that alter their phosphorylatable target sites.

a. Each kinase recorded in PhosphoAtlas was analyzed for its number of known, cognate heptameric peptide targets (x-axis) versus its mean number of total variants per HPR it phosphorylates (y-axis). Beyond identifying kinases whose substrate networks are the most alterable by cancer mutations, this spectrum also identifies kinases for which no variants are reported to alter their phosphorylation target sites (listed in the panel underneath the graph).

b-c. Tables with examples of how kinases can be affected by variants in the target sites they phosphorylate. Each panel in (c) corresponds to the selected set of kinases indicated in (a).

Supplement Figures S13-S14. Map of the tumor kinome.

We generated a projection of the cumulative incidence of all kinase and substrate nsHPRv's, indistinctively of their tissue or cancer origin, on the human phospho-reactome (i.e. all PhosphoAtlas

edges and nodes that define both kinase-substrate and kinase-substrate-target site phospho-catalytic interactions (total edges $n=4,758$); **Figure 2a**). The legend describes that a red disk indicates a protein that functions as kinase only; the diameter increases proportionally with the number of substrates it phosphorylates; a minimum size was chosen so that disks remain visible even when having a minimal number of interaction partners. A light blue disk indicates a protein that functions both as a kinase and as a substrate; the diameter increases proportionally to the sum of all of kinases phosphorylating it and all substrates it phosphorylates. A dark blue disk indicates a protein that functions as a substrate only; the diameter increases proportionally with the number of kinases phosphorylating it. A green dashed connector indicates a phosphorylation reaction has been identified but without any known target site. A green solid connector indicates an individual phosphorylation with a known target site and no known non-synonymous mutations within the immediate target site range (with the phosphorylatable amino acid residue and site location indicated in the middle of the connector). A dark pink connector indicates an individual phosphorylation with a known target site, and with one or more known non-synonymous mutation(s) within the immediate target site range; the stroke width increases with the number of tumor mutations reported for that site in COSMIC. A dark pink margin that surrounds each red, violet or blue disc, increases proportionally with the number of known non-synonymous mutations within the immediate range of any target site. In both **Figures S13** and **S14**, the protein size (node) reflects the number of reaction partners (i.e. 1st degree neighbors, including both kinases (upstream/inbound) and substrates (downstream/outbound)). The difference between **Figures S13** and **S14** is that, in **Figure S13** the protein labels also reflect the number of reaction partners, while in **Figure S14** protein labels reflect the number of variants (i.e. non-synonymous tumor variants affecting their endogenous target sites). The overlay of the functionally significant tumor mutations onto the global network of phospho-nodes highlights the specific molecular units that are most repeatedly affected. This representation visualizes how the collection of non-synonymous mutations recurrently impacts kinase circuits. This complex network delineates the most distinctive, actionable and possibly vulnerable, oncogenic kinase pathways of cancer. The differential mutational susceptibilities of the phospho-target regions of the human proteome reveal the most variable and disrupted kinase circuits found across 1000's of tumors catalogued in COSMIC. In complement to **Figure 6**, **Figures S13-S14** include all kinase and substrate proteins from PhosphoAtlas, including an additional set of 419 protein nodes that are not connected to the main interconnected network (section below the main 'hairball'). This lower panel includes kinase genes/proteins with unknown or unverified target substrates.

Supplement Figures S15-S16-S17. Examples of kinase and substrate sub-networks, and the differential impact of tumor mutations on their phospho-signaling circuits.

An assortment of protein sub-networks and non-synonymous mutations that affect their connectivity to other proteins, was established using the Cytoscape session that visually integrates the network data from PhosphoAtlas. Examples of kinases (and their upstream kinases and downstream substrates) are represented in **Figures S15-S16** (BRAF, KIT, EGFR, CHEK2, JAK2, RPS6KB1, WEE1, STK11, PKN2, FYN, GSK3B). Examples of substrates (and their upstream kinases) are represented in **Figure S17** (TP53, CTNNB1, PTEN, CDC25C, CTNNB1, RB1, APC, PLCG1, STAT3). Legend for both (a) and (b) is shown in (c) (similar legend as for **Figure S13-S14**; in each of these hubs, the edges between protein nodes are representative of the target site specific phosphorylation interaction wherever known, while the width of the purple edges reflects the number of non-synonymous variants in the heptamer region of a specific target site). The Cytoscape-derived networks intuitively display the potential relationship between mutations and local signaling topology as represented by hubs of interacting kinases and their substrates. The diversity of network topologies and the selective, differential mutational susceptibilities of target regions suggest how cancer-associated aberrations can specifically impact phospho-hubs.

Supplemental Information – Supplementary References

We would like to acknowledge that, due to the very large body of work produced in the field of phospho-proteomics and PPI networks, many more citations from important work are unfortunately missing from the references listed in this article.

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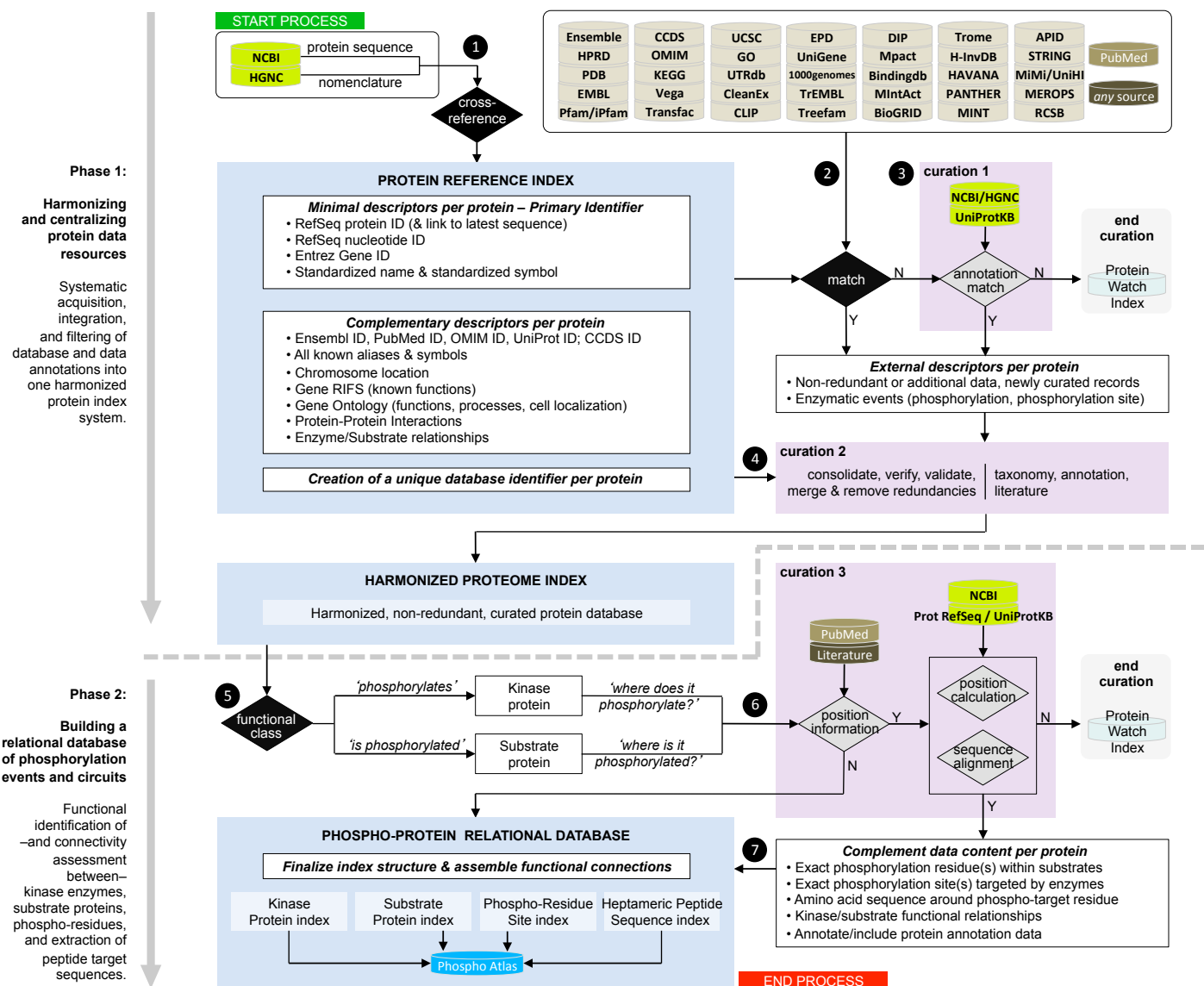
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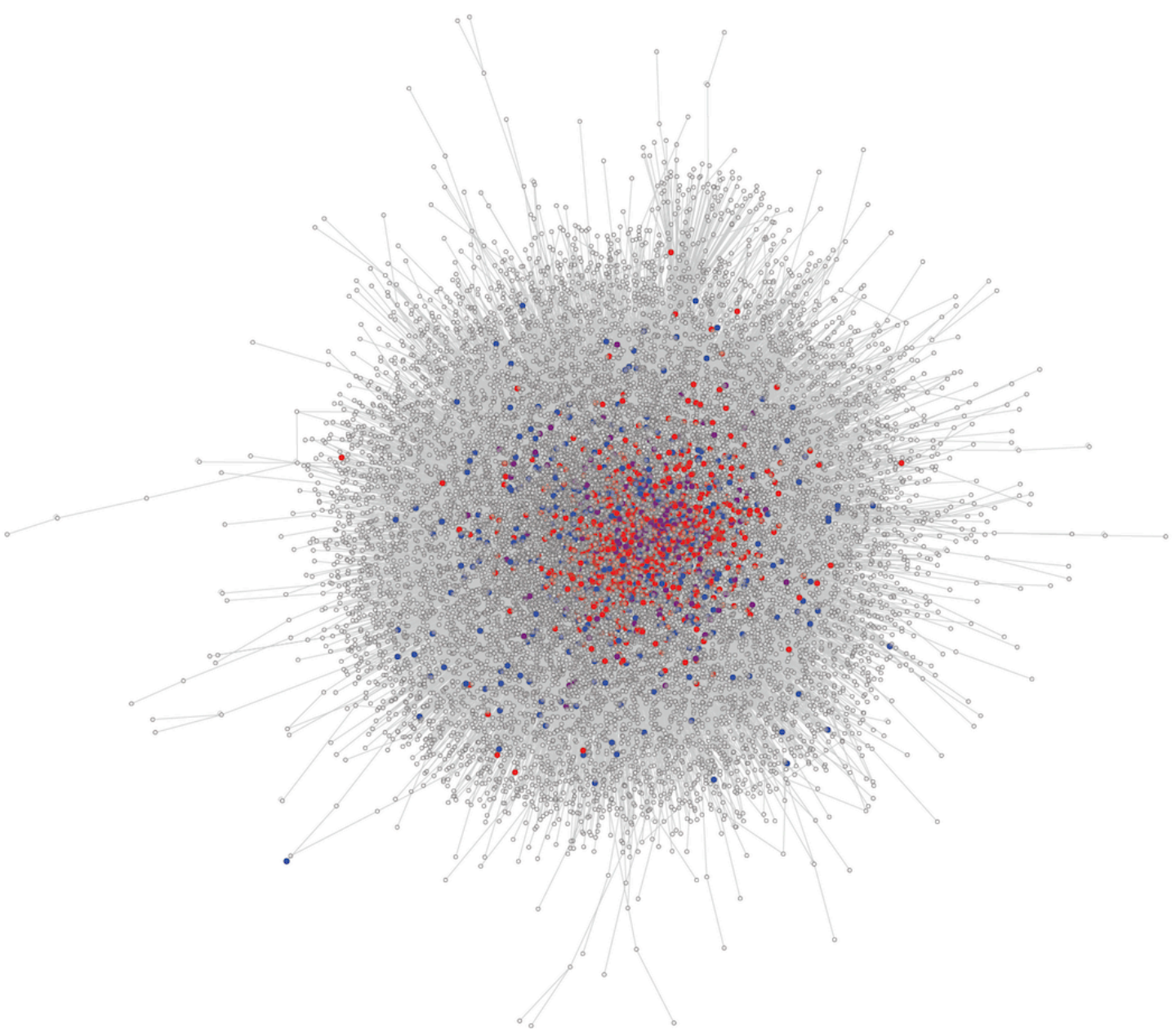
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**An atlas of the human kinome reveals the mutational
landscape underlying dysregulated phosphorylation
cascades in cancer**

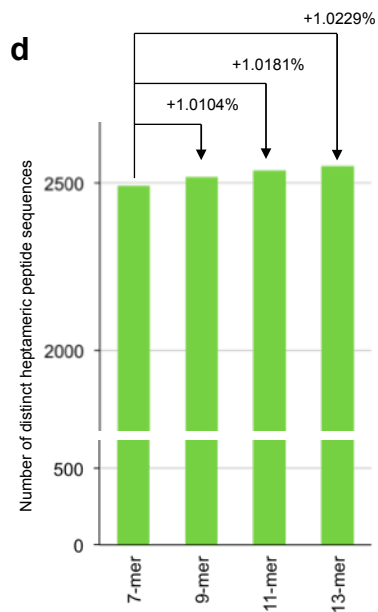
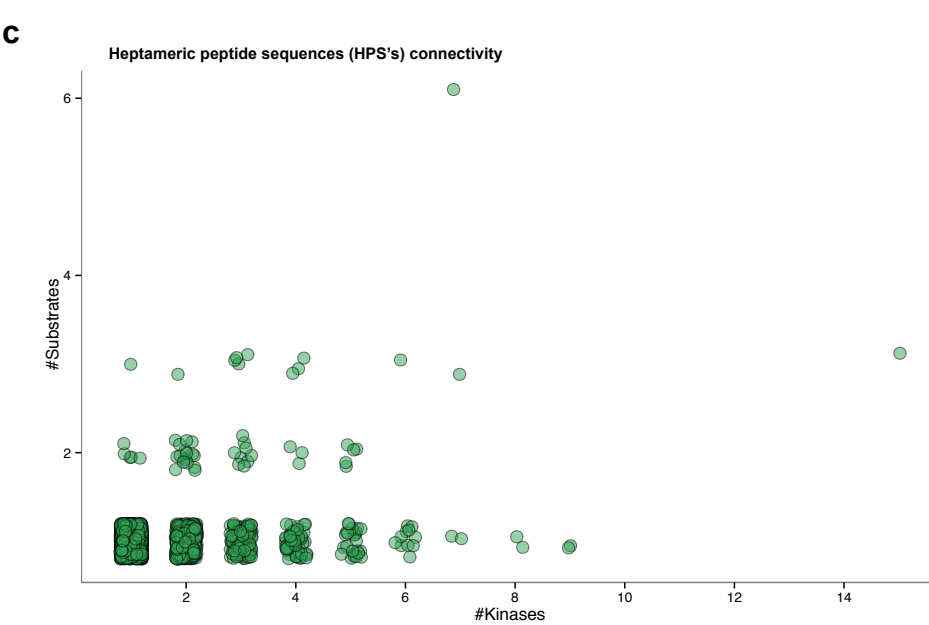
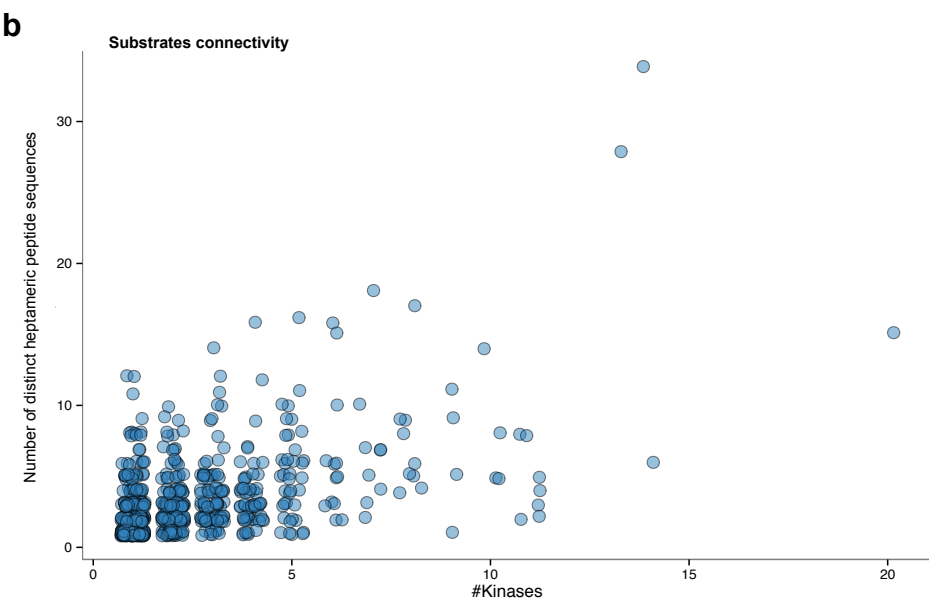
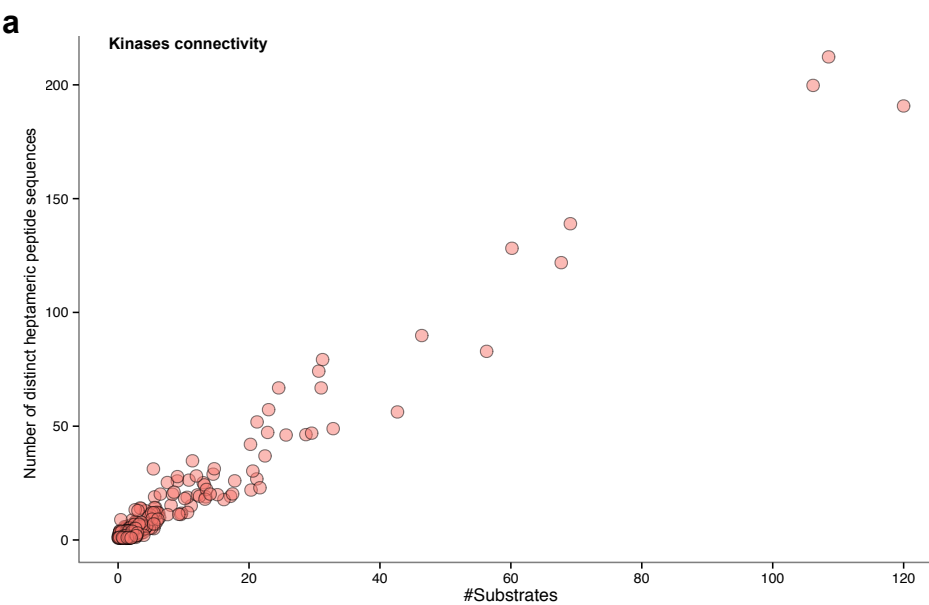
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Wolf, Christina Yau, Aleksandr Pankov, Evelyn Pei Rong Lee, Lamorna
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**SUPPLEMENTARY FIGURES
S1 to S17**

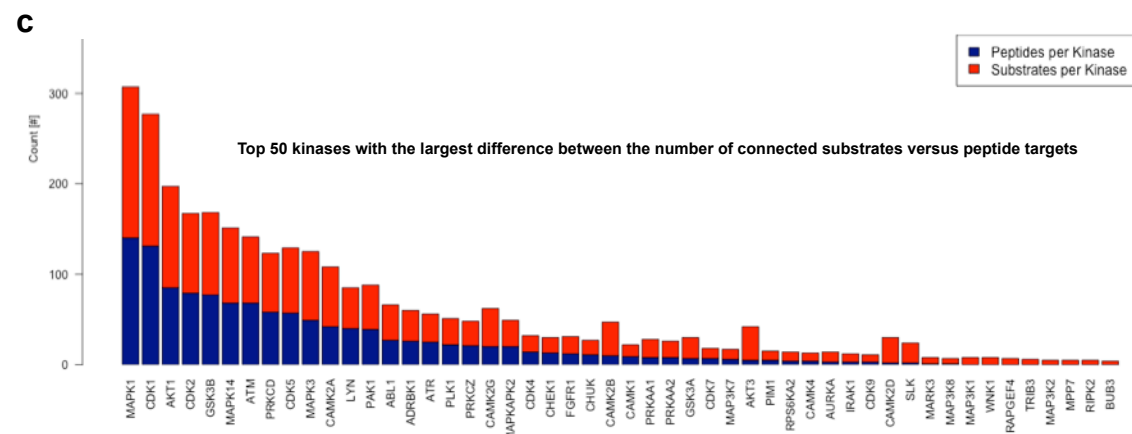




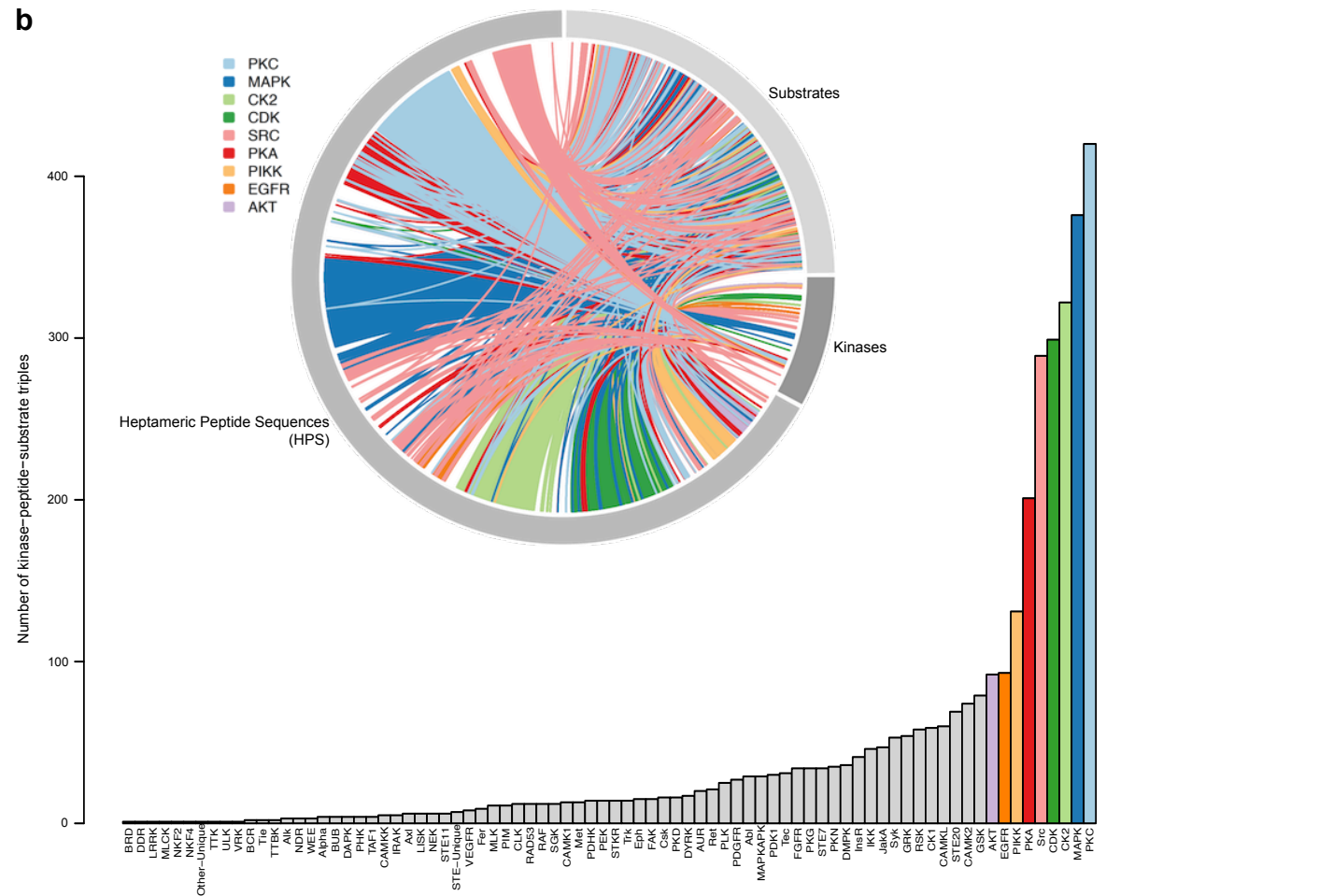
Supplement Figure S2



Supplement Figure S3



b

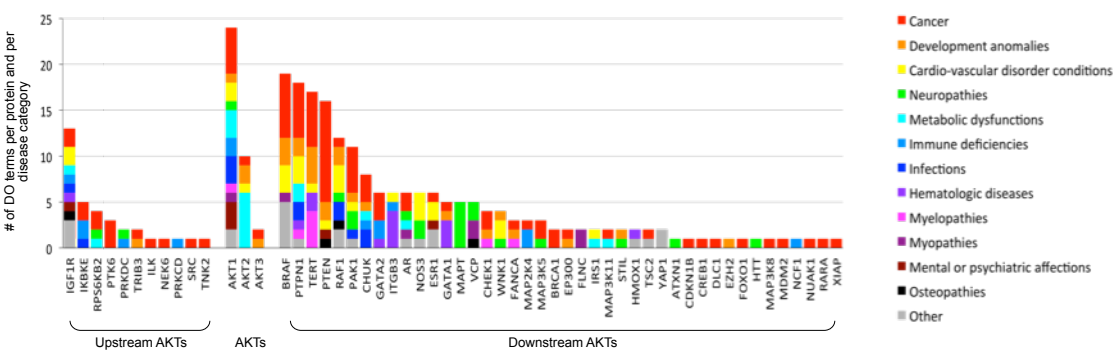


Supplement Figure S5

a

Disease category (Disease Ontology-terms were classified under 13 categories)	# of different DO terms per category	# of associated proteins per category	ontology details
Cancer	63	40	including malignant cancers, neoplasms, or hamartomas, across all tissue or cell types
Development anomalies	26	18	including congenital disorders, multisystem diseases, growth defects, mental retardation, intellectual disability
Cardio-vascular disorder conditions	23	14	including atherosclerosis, coronary artery disease, hypertension, cardiopathies
Neuropathies	18	14	including neurodegenerative disorders
Metabolic dysfunctions	11	9	including diabetes, obesity, insulin and glycemic dysregulation
Immune deficiencies	12	10	including auto-immune disorders
Infections	9	8	both viral & bacterial
Hematologic diseases	13	7	including anemic and bleeding disorders
Myelopathies	8	5	including bone marrow disorders
Myopathies	7	5	including muscular degenerative diseases
Mental or psychiatric affections	6	4	-
Osteopathies	4	4	-
Other	23	13	including disorders of eye, kidney, lung, skin, thyroid, digestive and reproductive system

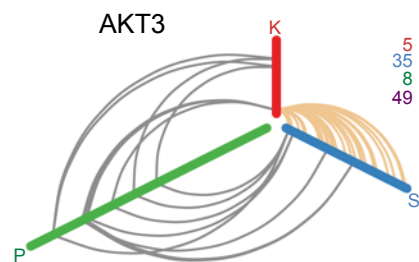
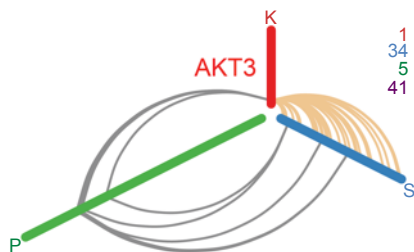
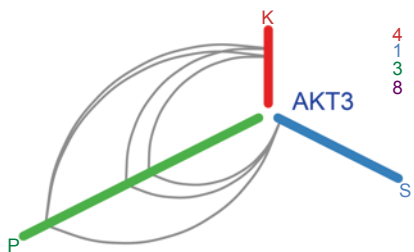
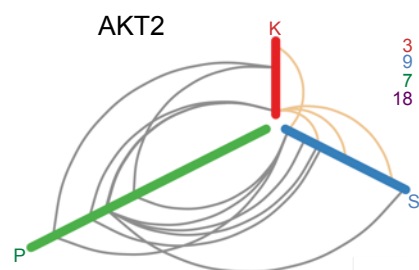
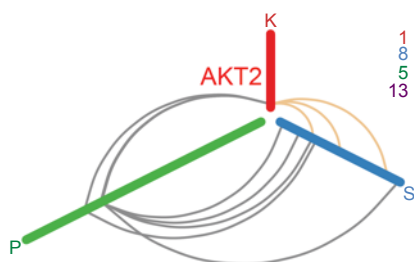
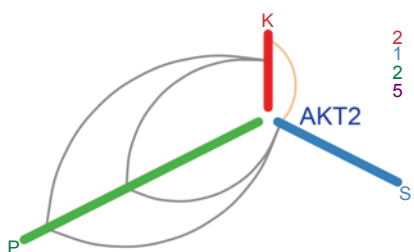
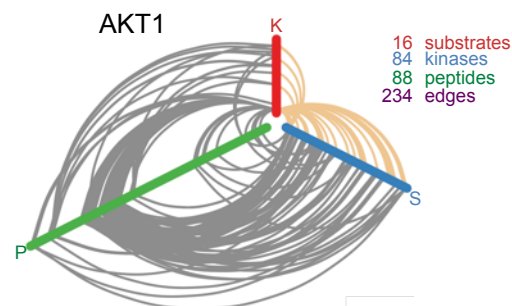
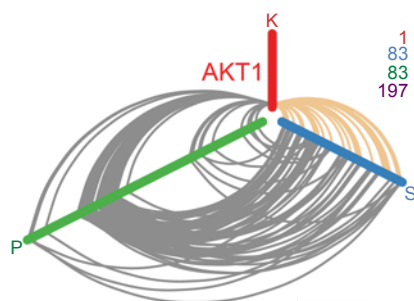
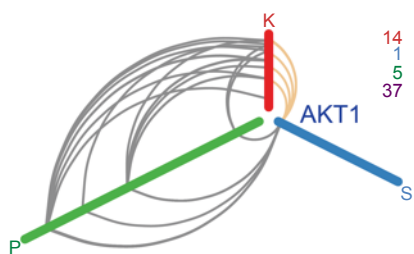
b



AKTs as substrate

AKTs as kinase

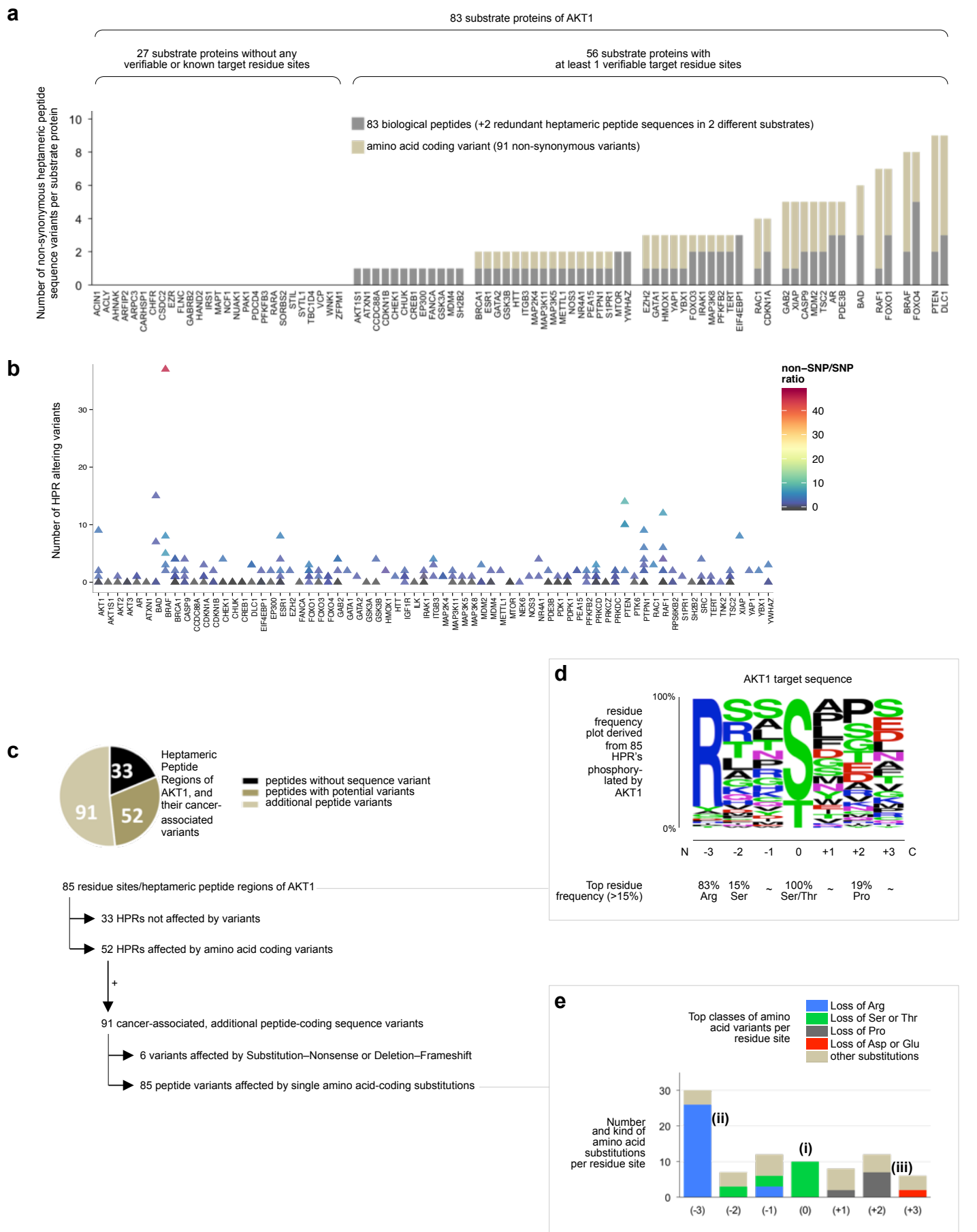
AKT-centric



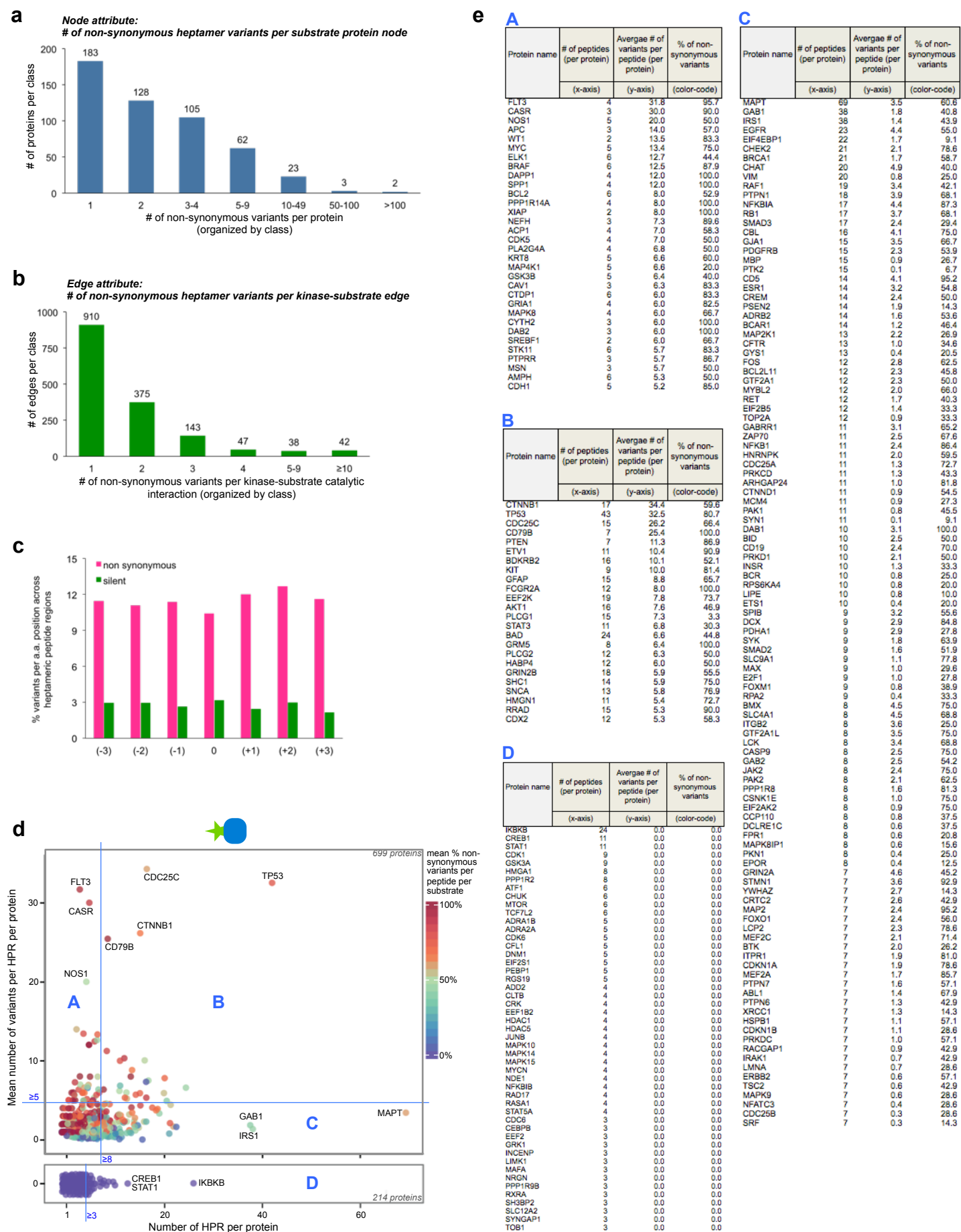
— Edges connecting a kinase to a substrate via a peptide target sequence

— Edges connecting kinase to substrates without a known/verified residue target site

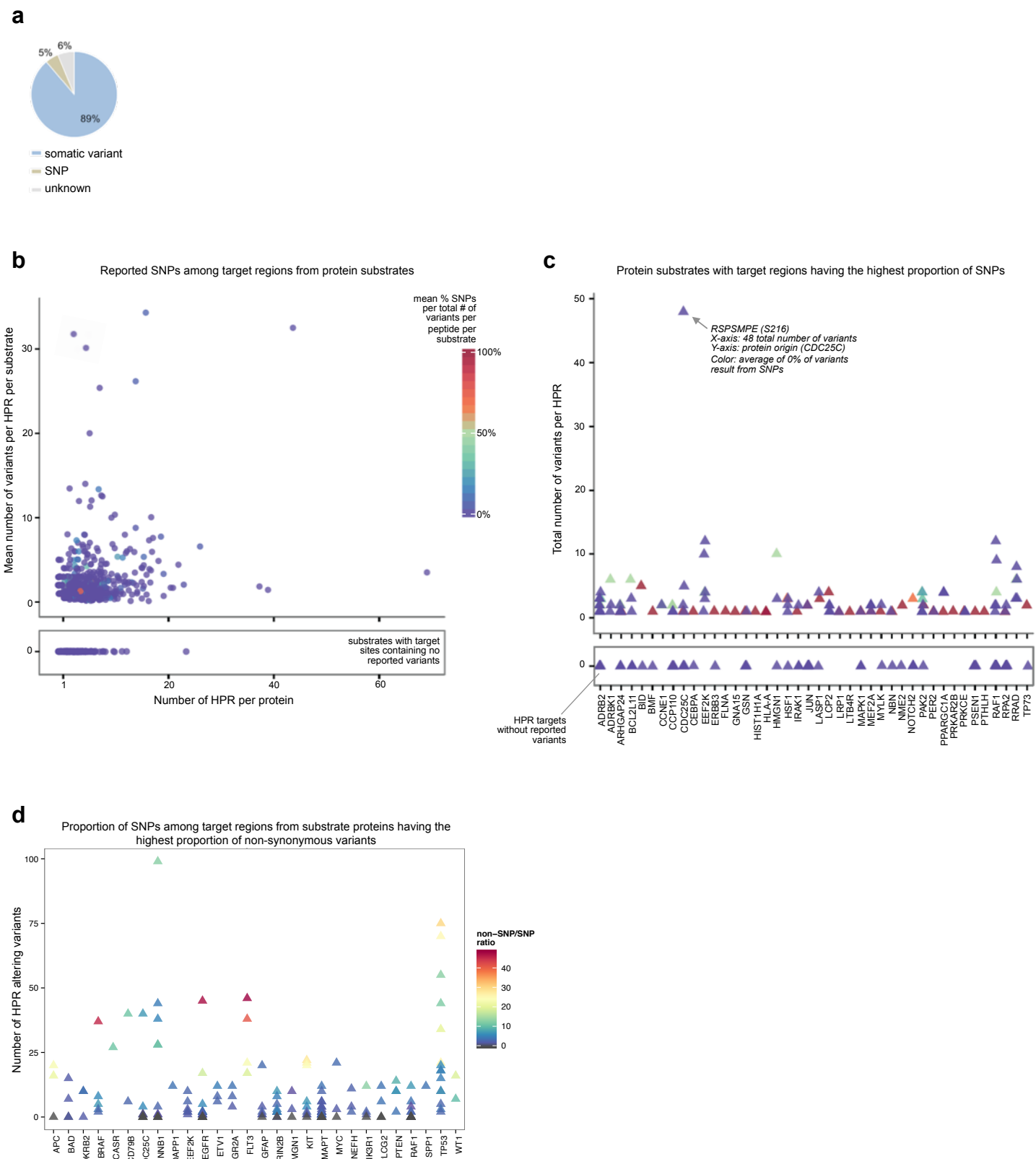
} total # of edges



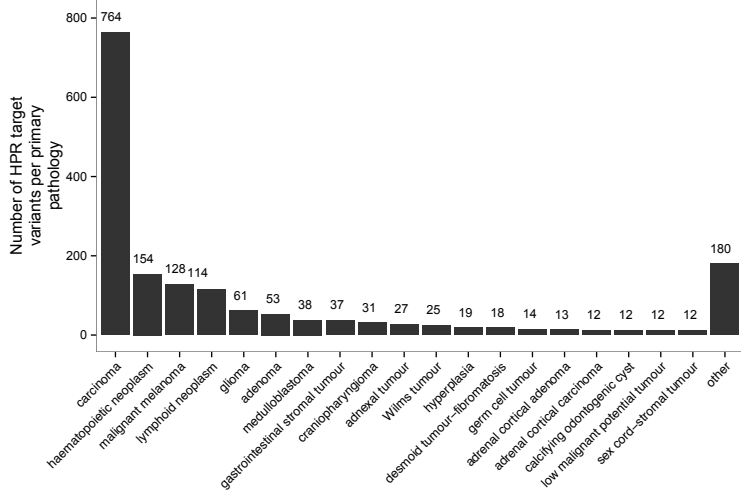
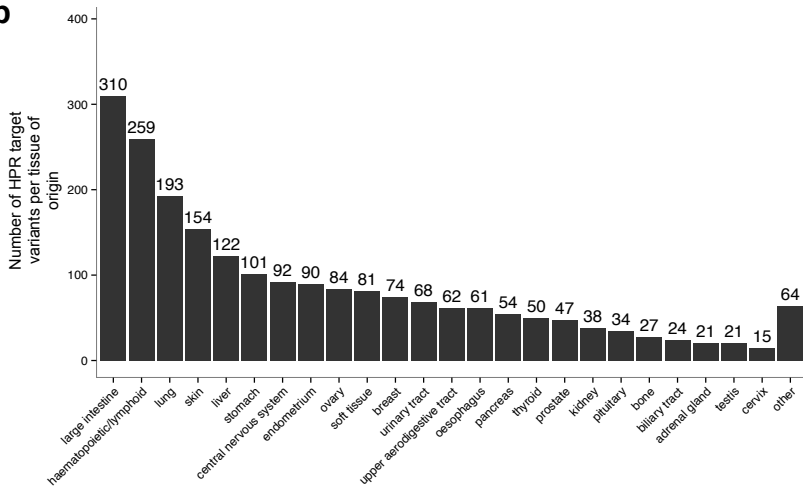
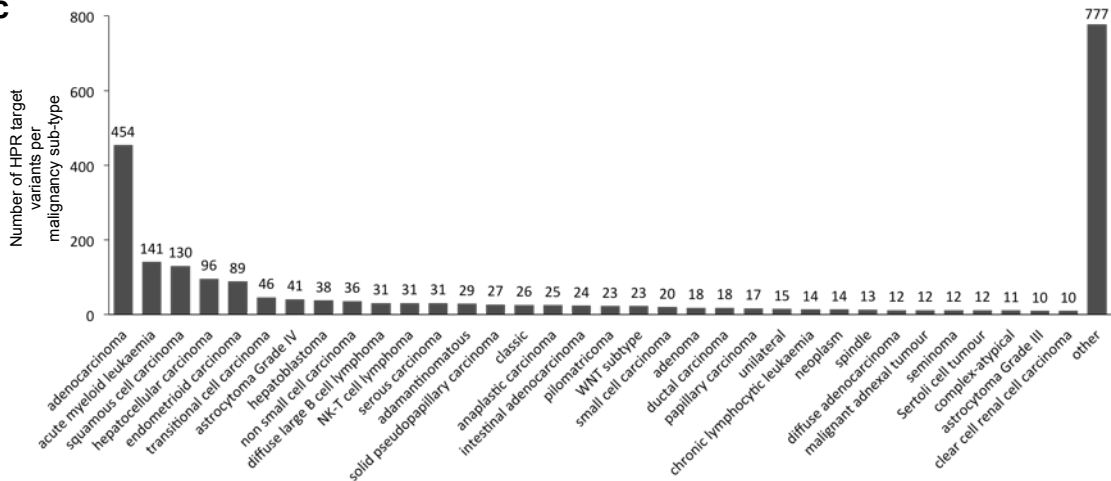
Supplement Figure S8

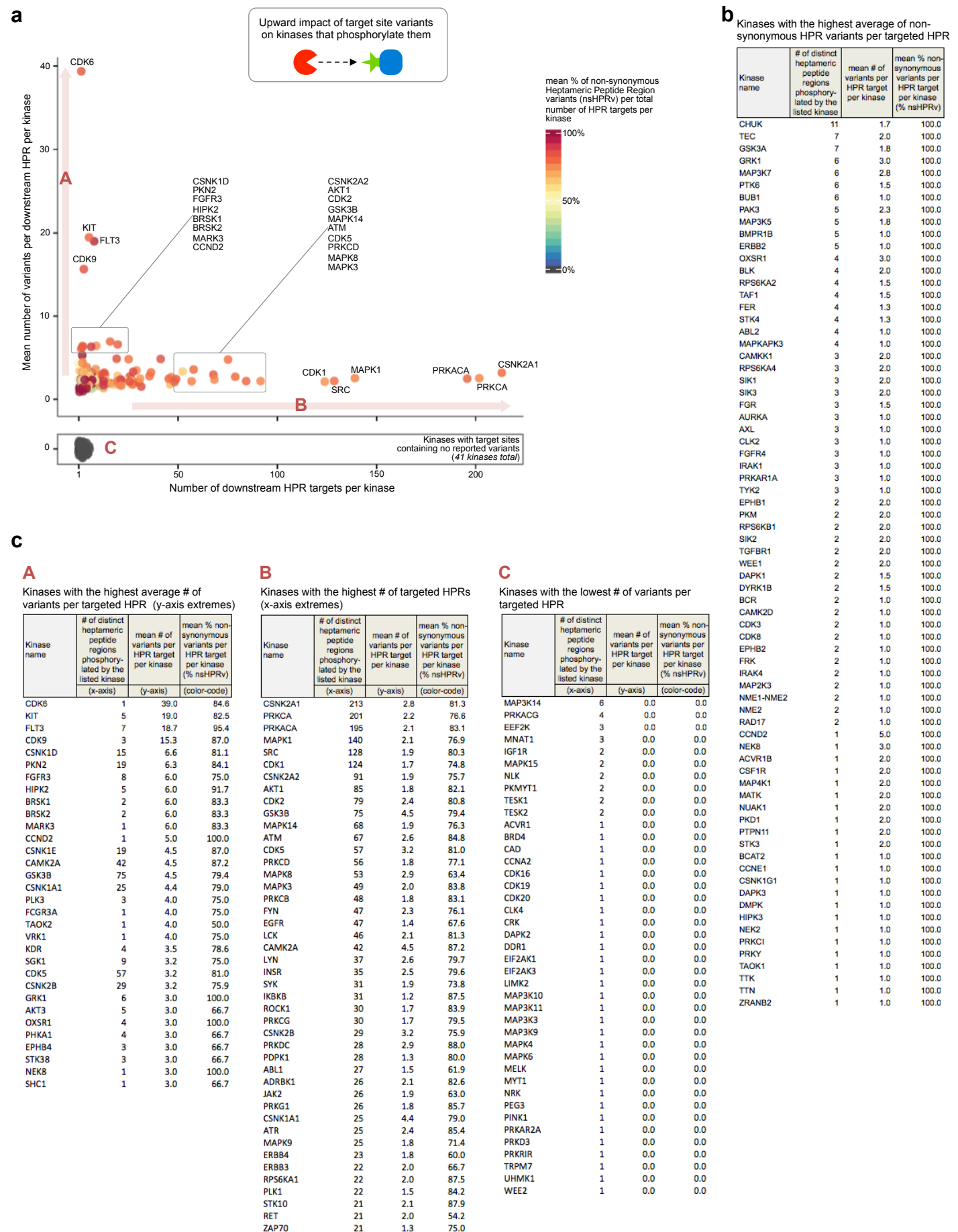


Supplement Figure S9

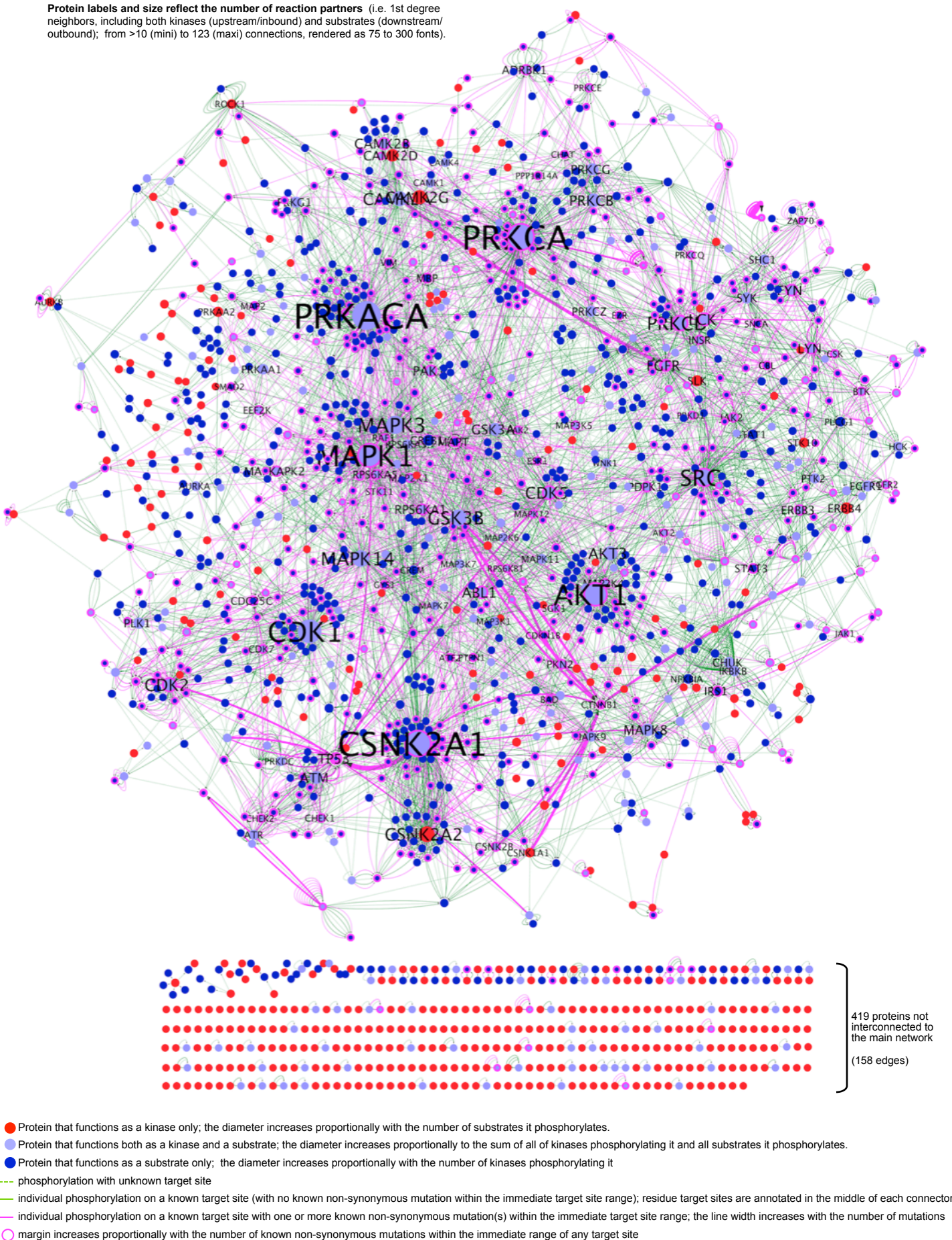


Supplement Figure S10

a**b****c**

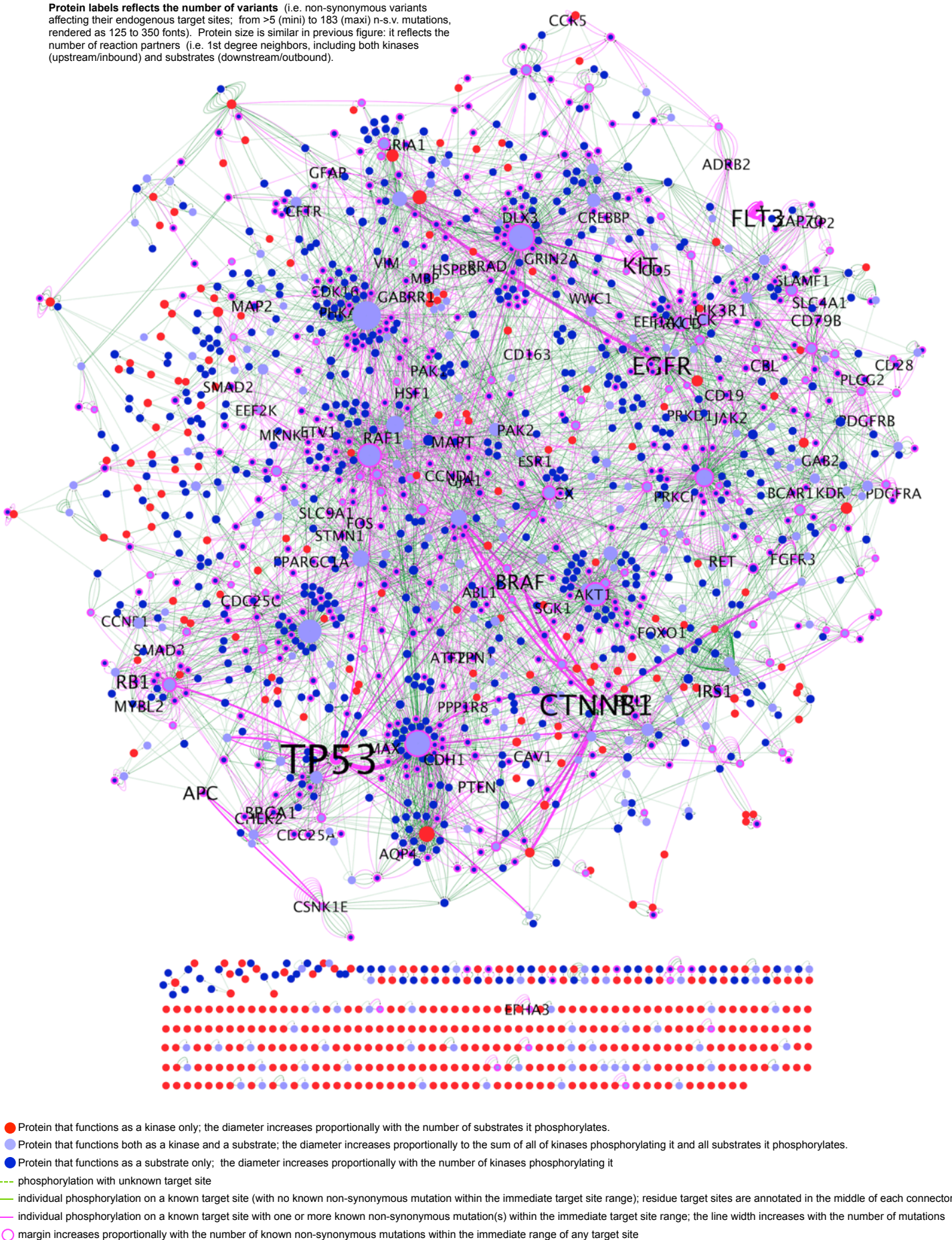


Protein labels and size reflect the number of reaction partners (i.e. 1st degree neighbors, including both kinases (upstream/inbound) and substrates (downstream/outbound); from >10 (mini) to 123 (maxi) connections, rendered as 75 to 300 fonts).

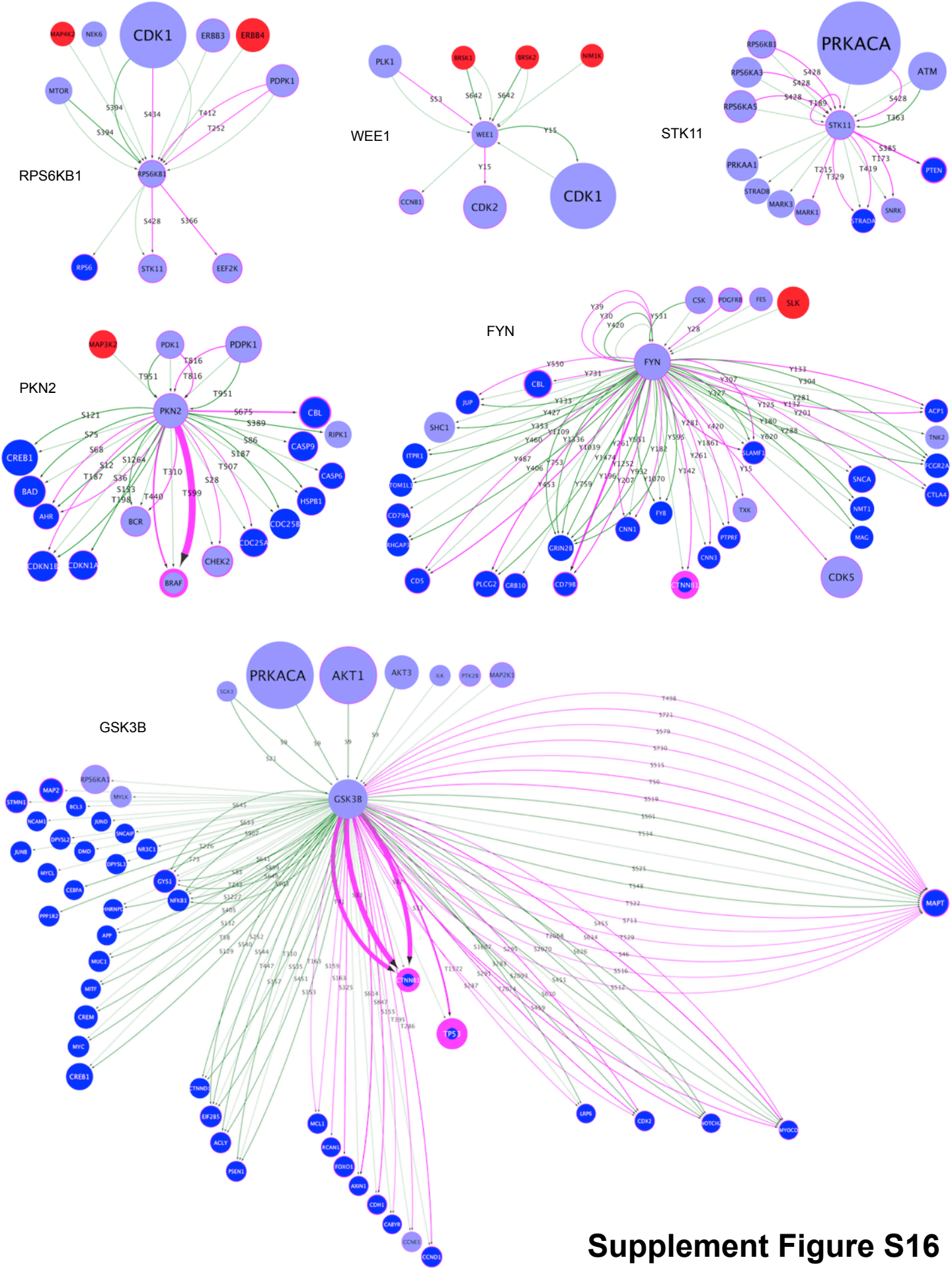


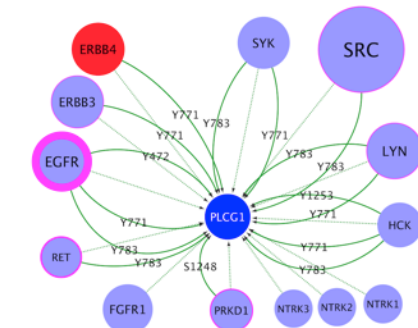
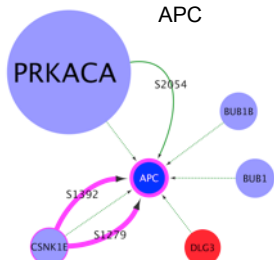
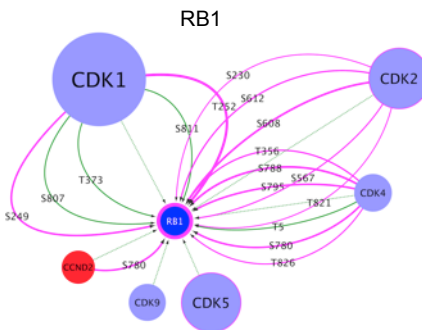
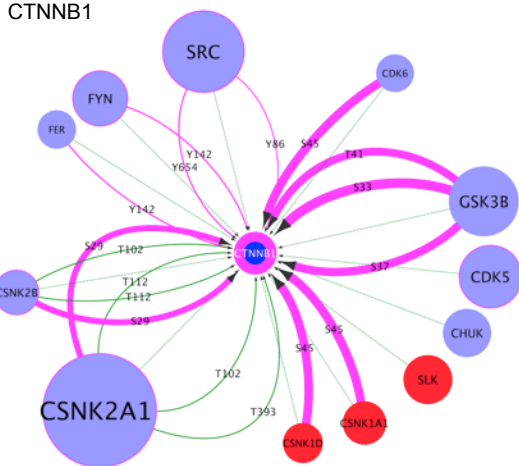
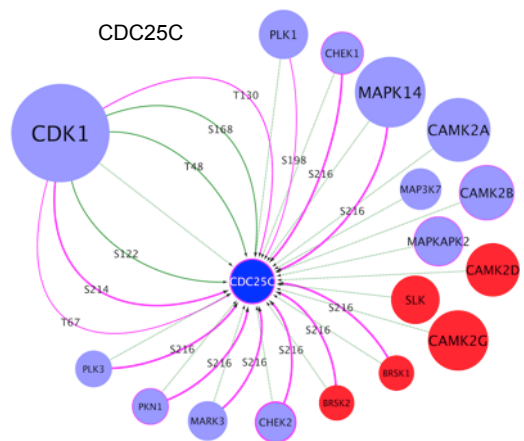
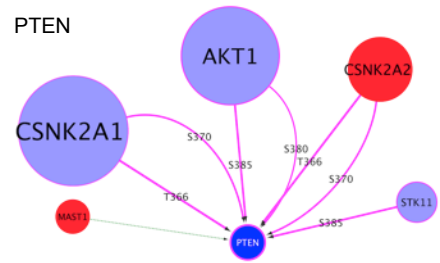
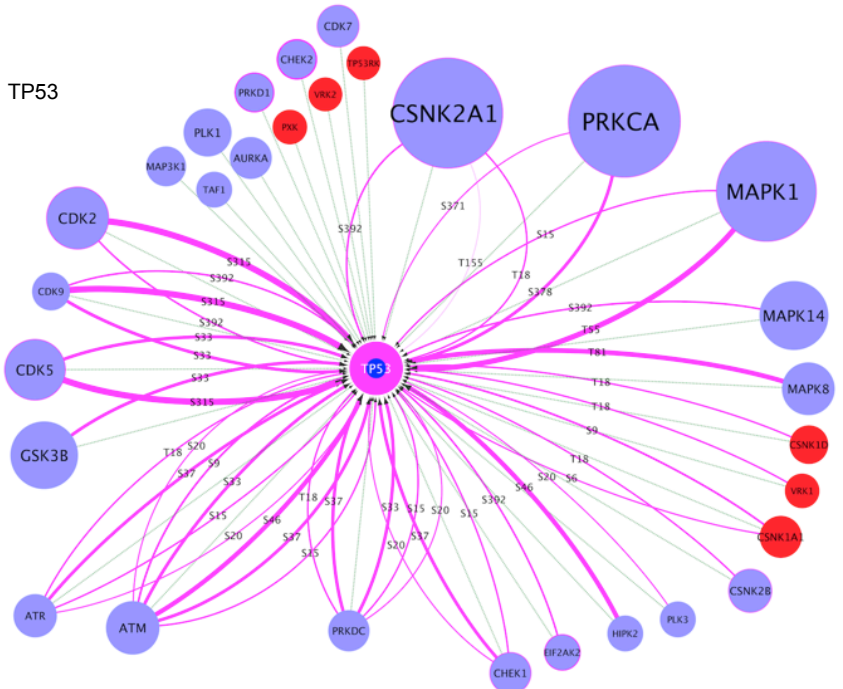
Supplement Figure S13

Protein labels reflect the number of variants (i.e. non-synonymous variants affecting their endogenous target sites; from >5 (mini) to 183 (maxi) n.s.v. mutations, rendered as 125 to 350 fonts). Protein size is similar in previous figure: it reflects the number of reaction partners (i.e. 1st degree neighbors, including both kinases (upstream/inbound) and substrates (downstream/outbound)).



Supplement Figure S14





Supplement Figure S17